PowerGuard AI

AI infrastructure forecasting and workload risk monitoring

Date range

2026/01/01 – 2026/01/14

Power risk threshold, kW

190

PowerGuard AI

Executive-grade AI infrastructure analytics for forecasting GPU power demand, detecting abnormal workload behavior, and improving visibility into energy cost, thermal risk, and compute utilization across modern AI workloads.

LLM Inference | Computer Vision Training | Batch Embeddings | GPU Telemetry | FastAPI + Streamlit

AVG POWER	PEAK POWER	14-DAY COST	RISK EVENTS	MODEL MAE
108.3 kW	272.8 kW	$4,490	97	4.85 kW
Mean GPU cluster draw	Highest observed load	Estimated energy spend	Thermal or power alerts	$R^2 = 0.97$

Executive Summary

PowerGuard AI identifies when AI workloads are likely to create energy-cost and cooling risk. Across the selected window, the platform observed an average utilization of **41.7%**, average temperature of **55.5°C**, and **97** risk events above the configured threshold. The forecasting model supports infrastructure planning by estimating short-term GPU power demand before workloads create expensive utilization spikes.

Business Problem

AI workloads create unpredictable power spikes that increase cloud/data-center cost and cooling risk.

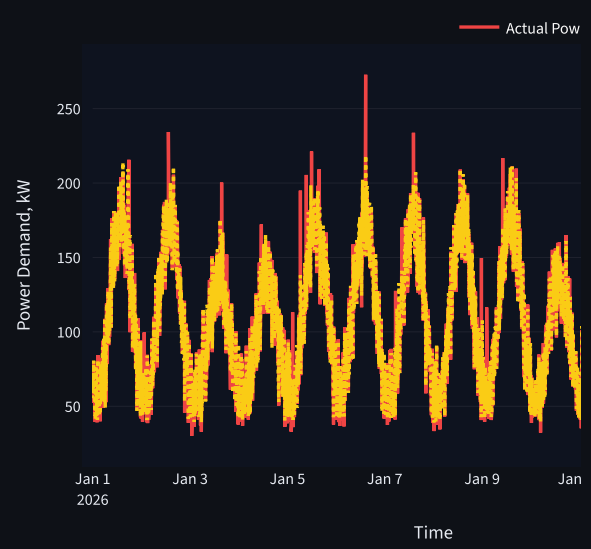
Analytics Solution

Forecast power demand using GPU utilization, temperature, memory pressure, and queue depth telemetry.

Operational Value

Help teams schedule workloads, detect anomalies, and make better infrastructure capacity decisions.

Power Forecast vs Actual Usage

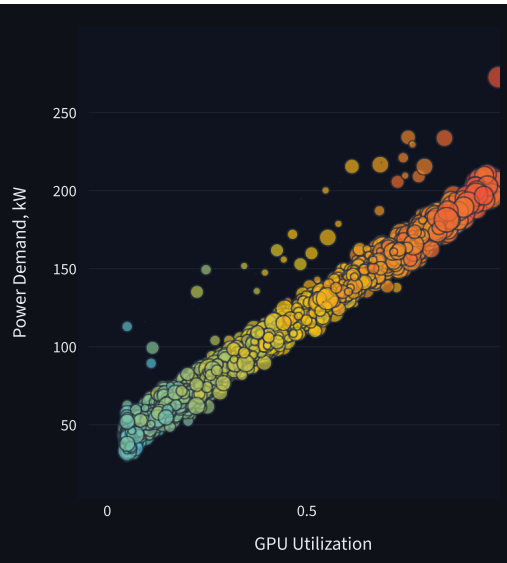


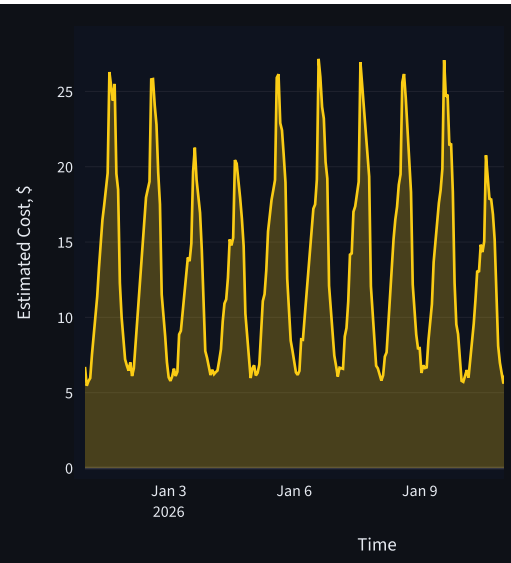
Workload Risk Mix

Workload type column not found in dataset.

GPU Utilization vs Power Demand

Hourly Cost Pattern





High-Risk Telemetry Events

timestamp	gpu_utilization	temperature_c	memory_utilization	active_jobs	queued_jobs
2026-01-01 10:25:00	0.8322	75.3186	0.649	57	4
2026-01-01 13:00:00	0.9373	73.2375	0.7395	64	8
2026-01-01 13:05:00	0.8279	75.1882	0.6744	44	6
2026-01-01 13:55:00	0.9517	73.3491	0.8203	56	4
2026-01-01 14:30:00	0.8191	76.4436	0.7063	54	3
2026-01-01 14:45:00	0.9096	76.1755	0.667	49	7
2026-01-01 14:55:00	0.98	82.5065	0.7659	65	7
2026-01-01 15:55:00	0.8254	76.2337	0.7593	48	6
2026-01-01 17:25:00	0.9773	78.0748	0.8638	63	7
2026-01-02 10:50:00	0.7808	75.5753	0.6056	51	0

PowerGuard AI is a portfolio project simulating AI data-center observability, forecasting, and infrastructure decision support.